

Model Question Paper-1 with effect from 2022-23 (CBCS Scheme)

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Fifth Semester B. E Degree Examination Environmental Studies

TIME: 01 Hours

Max.Marks:50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. For each question, after selecting your answer, **darken the appropriate circle** corresponding to the same question number on the **OMR** sheet.
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Answer the following Questions		*Bloom's Taxonomy Level	COs	Marks
1.	Habitat refers to a) Physical conditions of the place where organisms live b) Chemical conditions of the place where organisms live c) Both a & b d) None of a or b	L1	CO1	1
2.	Which of the following energy has the greatest potential among all the sources of renewable energy? a) Solar energy b) Wind Energy c) Thermal energy d) Hydro-electrical energy	L1	CO1	1
3.	The term 'Environment' has been derived from the French word which means to encircle or surround a) Environ b) Oikos c) geo d) Aqua	L1	CO1	1
4.	Which of the following statements is false? a) Inorganic nutrients are recycled in an ecosystem b) Energy 'flows' through the ecosystem in the form of carbon-carbon bonds c) Energy is recycled in an ecosystem d) Respiration process releases energy	L1	CO1	1
5.	In terms of greenhouse gas emissions, how good or bad is hydrogen fuel? a) Major contributor of greenhouse gas emissions b) Zero-emission fuel c) Lowest contributor of greenhouse gas emissions	L1	CO1	1

	d)Hydrogen cannot be used as fuel			
6.	A food web consists of a)a portion of a food chain b) an organism's position in a food chain c) interlocking food chains d) a set of similar consumers	L1	CO1	1
7.	In what form is solar energy is radiated from the sun? a)Ultraviolet Radiation b)Infrared radiation c)Electromagnetic waves d)Transverse waves	L1	CO1	1
8.	The by-product of the ocean thermal energy conversion is a)Hot water b)Desalinated water c)Chemicals d)Gases	L1	CO2	1
9.	Where can we find both running water as well as stagnant water? a)Marine ecosystems b)Wetlands c)Coral reefs d)Freshwater ecosystems	L1	CO2	1
10	----- is the conventional source for hydel power. a)Tidal wave b)Currents c)Water d)Ripples	L1	CO2	1
11	Ground water characteristics must be monitored at least once in a till design span. a) Week b) Month c) Day d) Quarter	L1	CO2	1
12	Acid rains are produced by (a)Excess NO ₂ and SO ₂ from burning fossil fuels (b)Excess production of NH ₃ by industry and coal gas (c)Excess release of carbon monoxide by incomplete combustion (d) Excess formation of CO ₂ by combustion and animal respiration.	L1	CO2	1
13	Which ecosystem is known as giant permanent pond? a)Lake Ecosystem b)Pond ecosystem c)Seashore ecosystem d)Marine ecosystem	L1	CO2	1
14	Which of the following is a producer in an ecosystem a)Plants and some bacteria capable of producing their own food b)Animals c) Human beings d) Fish	L1	CO2	1
15	The transfer of energy & nutrients from one feeding group of organization to another in a series is called. a)Energy chain b) Food Chain c) Balanced ecosystem d) Food Web	L1	CO1	1

16	Which one of the following is a result of climate change? a) Adequate rainfall b) Pure air c) Deficiency of freshwater d) Less soil pollution	L1	CO2	1
17	-----contaminant causes kidney and liver problems if present in groundwater. a) Benzene b) Toluene c) Benzotoulene d) Lead	L2	CO2	1
18	In an ecosystem, the flow of energy is a) Bidirectional b) Cyclic c) Unidirectional d) Multidirectional	L2	CO2	1
19	Which type of turbine is commonly used in tidal energy? a) Francis turbine b) Kaplan turbine c) Pelton wheel d) Gorlov turbine	L1	CO1	1
20	In an ecosystem biological cycling of materials is maintained by a) Producer b) Consumer c) Decomposer d) All of the above	L1	CO1	1
21	What type of energy is wind energy? a) Renewable energy b) Non-renewable energy c) Conventional energy d) Commercial energy	L1	CO1	1
22	Disaster management deals with situation that occurs after the disaster. a) True b) False	L1	CO1	1
23	What is the significance of the ionosphere? a) Aviation movements b) High frequency radio transmission c) Regulates weather d) All of the mentioned	L1	CO1	1
24	World Environment day is on a) 5th May b) 5th June c) 18th July d) 16th August	L2	CO3	1
25	How is hydrogen gas produced from fossil fuels? a) Partial oxidation of methane b) Electrolysis c) Evaporation d) Biomass gasification	L1	CO2	1
26	The Taj Mahal in India is affected by a) Fog b) Acid rain c) Water pollution d) Spoil Pollution	L1	CO1	1

27	Environmental Impact Assessment (EIA) is mandatory under which one of the following India legislations: a) Indian Forest Act b) Air (Prevention and Control of Pollution) Act c) Wildlife Protection Act d) Environment (Protection) Act	L2	CO3	1
28	What is the primary aim of the Air (Prevention and Control of Pollution) Act, 1981? a) To promote water conservation b) To control air pollution and preserve air quality c) To regulate waste management practices d) To protect soil quality	L2	CO3	1
29	Under the Solid Waste Management Rules, 2016, which of the following is NOT a duty of waste generators? a) To segregate waste at the source b) To manage waste through disposal in landfills c) To register with the authorities d) To ensure waste is handled by authorized persons	L1	CO3	1
30	Which year was the Water (Prevention and Control of Pollution) Act enacted? a) 1980 b) 1974 c) 1991 d) 1986	L2	CO3	1
31	Under the Solid Waste Management Rules, 2016, which of the following is NOT a duty of waste generators? a) To segregate waste at the source b) To manage waste through disposal in landfills c) To register with the authorities d) To ensure waste is handled by authorized persons	L2	CO3	1
32	The Water Act, 1974 led to the establishment of which regulatory body? a) Central Water Commission b) Central Pollution Control Board and State Pollution Control Boards c) Ministry of Environment and Forests d) National Green Tribunal	L2	CO3	1
33	According to the Biomedical Waste Management Rules, 2016, which color bin is designated for disposing of infectious waste? a) Yellow b) Blue c) Red d) Black	L1	CO2	1
34	What is the process of movement downwards of vadose water called? a) Infiltration b) Filtration c) Deposition d) Down-flow	L1	CO2	1
35	What is called for the movement of surface litter and topsoil from one place to another? a) Soil submerge b) Soil degradation c) Soil erosion d) Soil pollution	L1	CO1	1

36	-----is caused by drinking water high in nitrates. a) Cholera b) Methemoglobinemia c) Kidney problem d) Liver problem	L1	CO1	1
37	In which Act, noise is included as an environmental pollutant? a) 1974 b) 1981 c) 1988 d) 1994	L2	CO3	1
38	What is the upper surface of the zone saturation called? a) Aquifer b) Aquiclude c) Water table d) Aquifuge	L1	CO1	1
39	How many primary pollutants are there? a) Three b) Five c) Seven d) Nine	L1	CO1	1
40	The term 'Municipal Solid Waste' is used to describe which kind of solid waste? a) Hazardous b) Toxic c) Non hazardous d) Non toxic	L1	CO1	1
41	Open cycle OTEC uses surface water directly to make electricity. a) Hot b) Warm c) Cool d) Icy	L1	CO2	1
42	What does OTEC stand for? a) Ocean thermal energy cultivation b) Ocean thermal energy conversion c) Ocean techno energy conservation d) Ocean thermal energy consumption	L1	CO2	1
43	Which of the turbine can be mounted vertically and horizontally? a) Pelton wheel b) Kaplan turbine c) Gorlov turbine d) Francis turbine	L1	CO2	1
44	What type of energy is wind energy? a) Renewable energy b) Non-renewable energy c) Conventional energy d) Commercial energy	L1	CO2	1
45	Tsunami detectors are placed in sea at kms from shore. a) 25 b) 100 c) 50 d) 85	L2	CO1	1
46	What makes desert region to become highly unproductive? a) Salinity b) Sunlight c) Temperature	L1	CO1	1

	d) Increase in the rain			
47	How can desert ecosystems be conserved? a) By minimizing the human activity b) By pouring water to desert area c) By deforestation d) By killing organisms	L1	CO1	1
48	How the tropical rain forest gets the name? a) Due to less rain b) Due to heavy rain c) Due to moderate rain d) Due to no rain required	L1	CO1	1
49	How can we control acid rain which is causing due to the exhaust fumes on the atmosphere by cars? a) By burning more fuels b) By using old engine vehicles c) By using ignition d) By using catalytic converters	L1	CO1	1
50	Which is the most acidic in pH scale? a) 0 b) 7 c) 10 d) 14	L2	CO2	1

Model Question Paper-1/2 with effect from 2021 (CBCS Scheme)

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Model Question Paper-2 with effect from 2022-23 (CBCS Scheme)

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Fifth Semester B. E Degree Examination Environmental Studies

TIME: 01 Hours

Max.Marks:50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the fifty questions, each question carries one mark.
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Answer the following Questions		*Bloom's Taxonomy Level	COs	Marks
1.	Which type of ecosystem is characterized by high biodiversity, tall trees, and a warm climate year-round? a) Desert ecosystem b) Riverine ecosystem c) Forest ecosystem d) Oceanic ecosystem	L1	CO1	1
2.	Which of the following best defines an ecosystem? a) A community of organisms interacting with each other only b) A network of species sharing similar resources c) A system involving interactions between living organisms and their physical environment d) A group of species that live in the same geographic area	L1	CO1	1
3.	Which ecosystem is most adapted to extreme temperatures and low water availability? a) Wetlands b) Forest c) Desert d) Oceanic	L1	CO1	1
4.	What is the primary goal of the United Nations' Sustainable Development Goal (SDG) 1? a) Achieve universal education b) Eradicate poverty in all its forms c) Ensure clean water and sanitation d) Combat climate change	L1	CO1	1
5	What is the primary characteristic of wetlands? a) High salinity b) Frequent flooding and water saturation c) Sparse vegetation	L1	CO1	1

	d) Rocky terrain			
6.	SDG 13 focuses on which critical global issue? a) Marine life conservation b) Climate action c) Gender equality d) Sustainable agriculture	L1	CO1	1
7.	SDG 14 focuses on the conservation and sustainable use of: a) Freshwater resources b) Terrestrial ecosystems c) Oceans, seas, and marine resources d) Urban environments	L1	CO1	1
8.	Which of the following ecosystems plays a crucial role in carbon sequestration? a) Oceanic ecosystem b) Desert ecosystem c) Riverine ecosystem d) Mountain ecosystem	L1	CO2	1
9.	Which SDG promotes responsible consumption and production patterns? a) SDG 12 b) SDG 9 c) SDG 5 d) SDG 15	L1	CO2	1
10	Which of the following is the largest source of freshwater on Earth? a) Rivers b) Glaciers and ice caps c) Groundwater d) Lakes	L1	CO2	1
11	Which of the following is a primary indicator of water quality? a) Temperature b) Turbidity c) Salinity d) pH level	L1	CO2	1
12	Which of the following diseases is primarily water-borne? a) Tuberculosis b) Cholera c) Malaria d) Dengue	L1	CO2	1
13	What percentage of the Earth's water is freshwater? a) 97% b) 50% c) 2.5% d) 10%	L1	CO2	1
14	Which of the following organisms causes typhoid? a) Vibrio cholerae b) Salmonella typhi c) Plasmodium falciparum d) Escherichia coli	L1	CO2	1
15	Malaria is an example of which type of disease? a) Water-borne b) Air-borne c) Water-induced d) Food-borne	L1	CO1	1
16	The recommended safe limit of fluoride in drinking water by the	L1	CO2	1

	WHO is: a) 1.5 mg/L b) 2.5 mg/L c) 3.0 mg/L d) 0.5 mg/L			
17	Excess fluoride in drinking water primarily causes which health condition? a) Rickets b) Fluorosis c) Goiter d) Scurvy	L2	CO2	1
18	Which of the following is a non-conventional source of energy? a) Coal b) Natural Gas c) Solar Energy d) Petroleum	L2	CO2	1
19	Hydropower is considered a type of: a) Solar energy b) Non-renewable energy c) Renewable energy d) Geothermal energy	L1	CO1	1
20	Fluoride in drinking water affects which part of the human body the most? a) Liver b) Teeth c) Heart d) Lungs	L1	CO1	1
21	Which country is the largest producer of wind energy? a) India b) China c) Germany d) USA	L1	CO1	1
22	What is the primary advantage of using solar energy? a) Low installation cost b) Non-renewable source c) Zero greenhouse gas emissions d) High maintenance cost	L1	CO1	1
23	Hydrogen is considered a clean energy source because: a) It is abundant in fossil fuels b) It produces only water as a by-product c) It is cheaper than coal d) It can only be used in nuclear reactors	L1	CO1	1
24	Which form of energy is stored in fossil fuels? a) Nuclear Energy b) Thermal Energy c) Chemical Energy d) Electrical Energy	L2	CO3	1
25	Which gas is the primary contributor to the greenhouse effect? a) Oxygen b) Carbon dioxide c) Nitrogen d) Argon	L1	CO2	1
26	What is the primary effect of excessive nutrients like nitrogen and phosphorus in water bodies? a) Acidification	L1	CO1	1

	b) Eutrophication c) Ozone depletion d) Desertification			
27	Which of the following is a major source of water pollution? a) Agricultural runoff b) Solar energy production c) Wind energy turbines d) Geothermal power plants	L2	CO3	1
28	Which pollutant is most commonly associated with industrial wastewater? a) Pesticides b) Heavy metals c) Carbon dioxide d) Methane	L2	CO3	1
29	What is the unit used to measure noise levels? a) Hertz (Hz) b) Decibel (dB) c) Joule (J) d) Pascal (Pa)	L1	CO3	1
30	Which of the following is a major source of air pollution in urban areas? a) Agricultural activities b) Automobile emissions c) Deforestation d) Solar power plants	L2	CO3	1
31	What is a key preventive measure for reducing air pollution? a) Reducing deforestation b) Using renewable energy sources c) Increasing the use of fossil fuels d) Promoting single-use plastics	L2	CO3	1
32	Which pollutant is responsible for causing acid rain? a) Carbon monoxide b) Sulfur dioxide c) Methane d) Ammonia	L2	CO3	1
33	Which health condition is most commonly associated with air pollution? a) Osteoporosis b) Respiratory diseases c) Skin infections d) Digestive disorders	L1	CO2	1
34	Which type of waste is classified under biomedical waste? a) Construction debris b) Used syringes c) E-waste d) Food packaging	L1	CO2	1
35	Which of the following is a major source of solid waste in urban areas? a) Agriculture b) Industries c) Households d) Forests	L1	CO1	1
36	Which type of solid waste is categorized as biodegradable? a) Glass bottles b) Plastic bags c) Food scraps	L1	CO1	1

	d) Aluminum cans			
37	What is the final step in the functional elements of solid waste management? a) Waste collection b) Waste transportation c) Waste disposal d) Waste generation	L2	CO3	1
38	What is a key characteristic of biomedical waste? a) Non-hazardous b) Radioactive c) Infectious and hazardous d) Biodegradable and harmless	L1	CO1	1
39	According to the Solid Waste Management Rules, 2016, which of the following is a duty of waste generators? a) Ensure segregation of waste at source b) Transport waste to landfill sites c) Burn the waste in open spaces d) Dispose of waste in water bodies	L1	CO1	1
40	Under the Biomedical Waste Management Rules, 2016, biomedical waste should be treated and disposed of within how many hours of generation? a) 12 hours b) 24 hours c) 48 hours d) 72 hours	L1	CO1	1
41	Which hazardous pollutant is commonly found in e-waste? a) Mercury b) Methane c) Sulfur dioxide d) Carbon monoxide	L1	CO2	1
42	Which of the following is a basic principle of e-waste management? a) Open burning of e-waste b) Extended Producer Responsibility (EPR) c) Dumping in water bodies d) Burying without treatment	L1	CO2	1
43	The E-Waste (Management) Rules, 2022, introduced stricter guidelines on: a) Banning e-waste recycling b) EPR targets for producers c) Eliminating e-waste management entirely d) Exporting all e-waste overseas	L1	CO2	1
44	What is the primary aim of the E-Waste (Management) Rules, 2022? a) Promote informal recycling of e-waste b) Encourage scientific and safe disposal of e-waste c) Increase landfilling of e-waste d) Restrict the use of recycled components	L1	CO2	1
45	According to the E-Waste (Management and Handling) Rules, 2011, who is primarily responsible for the collection of e-waste? a) Municipal corporations b) Producers and manufacturers c) Consumers d) Educational institutions	L2	CO1	1
46	Which of the following environmental issues is caused by improper e-waste disposal?	L1	CO1	1

	a) Ozone depletion b) Soil and groundwater contamination c) Desertification d) Ocean acidification			
47	Which of the following devices contributes the most to global e-waste generation? a) Refrigerators b) Mobile phones and computers c) Solar panels d) Wind turbines	L1	CO1	1
48	Which toxic substance in e-waste is known to cause kidney damage when improperly disposed of? a) Cadmium b) Carbon dioxide c) Nitrogen d) Phosphorus	L1	CO1	1
49	Which component of e-waste is most valuable for recycling due to its precious metal content? a) Plastic casings b) Circuit boards c) Batteries d) Cathode ray tubes	L1	CO1	1
50	Lead, often present in e-waste, primarily affects which human organ? a) Lungs b) Brain c) Liver d) Heart	L2	CO2	1

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Question Paper Version : A

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies and E – Waste Management

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
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1. Which of the following conceptual spheres of the environment is having the least storage capacity for matter?
a) Atmosphere b) Lithosphere c) Hydrosphere d) Biosphere
2. The ratio between energy flows at different points in a food chain is known as
a) Ecological capacity b) Ecological efficiency
c) Ecological assimilation d) Ecological potential
3. A predator is
a) An animal that is fed upon another animal
b) Animal that feeds upon both plants and animals
c) An animal that feeds upon another animal
d) A primary consumer
4. Why Rann of Kutch attracts aquatic birds in monsoon season?
a) Because desert land is converted to forest land
b) Because desert land is converted to snow
c) Because desert land do not convert
d) Because desert land is converted to salt marshes
5. Which kind of soil we can find on the surface of Thar desert?
a) Rocky b) Moist c) Fertile d) Aeolian
6. Which of the following type of forest important for watersheds?
a) Tropical Evergreen forests b) Tropical Deciduous forests
c) Tropical Montana forests d) Grassland forest

7. Hot spots areas have
 - a) Low density of biodiversity
 - b) Only endangered plants
 - c) High density of hot springs
 - d) High density of biodiversity
8. Sustainable Development means
 - a) meeting present needs without compromising on future needs
 - b) progress of human beings
 - c) balance between human needs and the ability of earth to provide the resources
 - d) all of these
9. The term Alpha diversity refers to
 - a) Genetic diversity
 - b) Community and ecosystem diversity
 - c) Species diversity within a community or ecosystem
 - d) Diversity among the plant
10. Algae , green plants and photosynthetic bacteria are
 - a) Autotrophic
 - b) Heterotrophic
 - c) Decomposers
 - d) Consumers
11. Veld type grasslands are located at
 - a) South Africa
 - b) South America
 - c) Australia
 - d) Britain
12. Which pyramid is always upright?
 - a) Energy
 - b) Biomass
 - c) Numbers
 - d) Food chain
13. In what form is solar energy is radiated from the Sun?
 - a) Ultraviolet Radiation
 - b) Infrared Radiation
 - c) Electromagnetic waves
 - d) Transverse waves
14. What does MHD stands for in the energy field?
 - a) Magneto Hydro Dynamic
 - b) Metal Hydrogen Detox
 - c) Micro Hybrid Drive
 - d) Metering Head Differential
15. The 'Miracle Material' that can turn CO₂ into liquid fuel is :
 - a) Propane
 - b) Copper
 - c) Graphene
 - d) Potassium
16. A tide whose difference between high and low tide is greatest.
 - a) Diurnal tide
 - b) Neap tide
 - c) Spring tide
 - d) Ebb tide
17. Which of the turbine can be mounted vertically and horizontally.
 - a) Pelton wheel
 - b) Kaplan turbine
 - c) Gorlov turbine
 - d) Francis turbine
18. Which type of fuel is removed from the reactor core after reaching end of core life service?
 - a) Burnt fuel
 - b) Spent fuel
 - c) Engine oil
 - d) Radioactive fuel
19. What is a fuel cell?
 - a) Converts heat energy to chemical energy
 - b) Converts heat energy to electrical energy
 - c) Converts chemical energy to electrical energy
 - d) Converts kinetic energy to heat energy

20. Which one of the following is the apex organization in our country in the field of pollution control?
 a) Water Pollution Control Board b) State Pollution Control Board
 c) Central Pollution Control Board d) Air Pollution Control Board
21. _____ is caused by drinking water high in nitrates.
 a) Cholera b) Kidney problem
 c) Liver problem d) Methemoglobinemia
22. Bhopal gas tragedy took place in the year _____ and the gas responsible was _____
 a) 1964, Hydrogen fluoride b) 1974, Methyl chloride
 c) 1984, methyl ISO – cyanide d) 1994, Methyl sulphate
23. The major chemical pollutants in photochemical smog are
 a) NO, NO₂, VOC, O₃, PAN b) N₂O, NO₂, VOC, O₃, PAN
 c) NO, NO₂, VOC, O₂, PAN d) NO, N₂O₅, VOC, O₃, PAN
24. The international protocol to protect the Ozone layer is
 a) Vienna protocol b) Kyoto protocol
 c) Cartagena protocol d) Montreal protocol
25. Which is the best and the worst method of plume behavior for pollution dispersion?
 a) Lofting and fumigation b) Trapping and fanning
 c) Coning and fumigation d) Fanning and Lofting
26. What is called for a Temporary hearing loss?
 a) Temporary ear pain b) Temporary hearing problem
 c) Temporary threshold shift d) Temporary hearing shift
27. What timings loud speakers shouldn't use in public areas?
 a) 10.00 pm to 5.00 am b) 11.00 pm to 6.00 am
 c) 1.00 am to 7.00 am d) 10.00 pm to 6.00 am
28. In which section, if a person violates the noise pollution regulations, is liable for penalty according to Environmental Protection Act, 1986.
 a) Section 12 b) Section 15 c) Section 18 d) Section 19
29. 'Minamata Disease' is caused due to
 a) Lead b) Arsenic c) Mercury d) Cadmium
30. The process of reducing the fluoride content from water is called as
 a) Chlorination b) Fluoridation
 c) Defluoridation d) Fanning and Lofting
31. What is the Dissolved oxygen value required for the survival of aquatic species?
 a) 7 mg/L b) 8.2 mg/L c) 6.5 mg/L d) 4 mg/L
32. Which among the following is used to dump the waste collected in the cities?
 a) Land fills b) Ocean c) River d) All of these

33. Which type of waste includes items such as leftover food, fruit peels and yard trimmings?

a) Hazardous waste b) Organic waste
c) Bio – medical waste d) Electronic waste
34. Which of the integrated waste management is reduced on an individual level?

a) Source Reduction b) Recycling c) Disposal d) Burning
35. What is called for the process of burning municipal solid waste in a properly designed furnace under suitable temperature and operating conditions?

a) Landfill b) Recycling c) Vermicomposting d) Incineration
36. The process of decomposition of biodegradable solid waste by earthworms is called

a) Landfill b) Vermicomposting c) Composting d) Shredding
37. _____ is a liquid that passes through solid waste and extracts suspended impurities from it

a) Leachate b) Sludge c) Distilled water d) Municipal
38. The colour code of plastic bag for disposing of microbial laboratory culture waste

a) black b) red c) blue d) white
39. Average hospital waste produced per bed per day in Government hospital is

a) 1.5 to 2 kg b) 0.5 – 4 kg c) 0.5 to 1 kg d) 0.5 – 2 kg
40. Which of the following are the main contributors of the e – waste in the world?

I. Refrigerators / freezers , washing machines , dishwashers.
II. Small household appliances
III. Personal computers, telephones , laptops , printers.
IV. Gas cylinders, chimneys and home appliances

a) Only I, II, III b) Only I & II c) Only I, III , IV d) All of these
41. What is Extended Producer Responsibility (EPR) as per the e – waste management rules in India?

a) The responsibility of consumer to manage e – waste
b) The responsibility of manufactures to manage e – waste throughout the product life cycle
c) The responsibility of retailers to manage e – waste disposal
d) The responsibility of informal recyclers to manage e – waste.
42. Which international agreement regulates the transboundary movements of hazardous waste, including e – waste?

a) Kyoto Protocol b) Paris agreement
c) Montreal Protocol d) Basel convention
43. Which colour bin is used for e – waste?

a) Blue b) Green c) Yellow d) Black
44. What are the health hazards which can be caused by E – waste?

a) Lung cancer b) DNA damage c) Brain d) All of these

45. Preparation of Guidelines for Environmentally sound Management of e – waste is a duty assigned to
a) Producer b) Consumer c) MOEFCC d) SP CB/PCC
46. What is India's global rank in e – waste?
a) 3 b) 13 c) 23 d) 33
47. When did the Karnataka State Pollution Control Board for prevention and control of water pollution constituted?
a) 1974 b) 1978 c) 1982 d) 1986
48. Aerosol consisting of liquid droplets is called as
a) Mist b) Dust c) Fog d) Aerosol
49. Which of the following is non – point source of water pollution?
a) Factories b) Sewage treatment plant
c) Urban and suburban land d) All of these
50. When is World Water day celebrated?
a) January 26th b) June 5th c) September 22nd d) March 22nd

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Question Paper Version : A

Fifth Semester B.E./B.Tech. Degree Examination, June/July 2025
Environmental Studies and E – Waste Management

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

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1. In an ecosystem biological cycling of materials is maintained by
a) Producer b) Consumer c) Decomposer d) All of these
2. The primary producers in a forest ecosystem are
a) Chlorophyll containing trees and plants b) Herbivores
c) Carnivores d) Bacteria and other micro-organisms
3. Which of the following is the terrestrial ecosystem?
a) Forest b) Grass land c) Desert d) All of these
4. Food web consists of
a) a portion of food chain b) an organism position in a food chain
c) interlocking food chain d) a set of similar consumers
5. The area of National Parks range between
a) 0.61 to 7818 kms b) 0.04 to 3162 kms c) 0.14 to 3612 kms d) 0.16 to 8718 kms
6. Which Indian state has its maximum area under the forest cover?
a) Maharashtra b) Madhya Pradesh c) Arunachal Pradesh d) Karnataka
7. How many parts are there in forest ecosystem?
a) Biotic type b) Abiotic type c) Both (a) & (b) d) None of these
8. To which of the following is Ramsar convention related?
a) Protection of wild life b) Protection of environment
c) Protection of rivers d) Protection of wetlands

9. Which of the following option is not included in the sustainable development parameters?
 - a) Gender inequality
 - b) Intergenerational and intragenerational equity
 - c) Growing annually
 - d) None of these
10. How many sustainable development goals are there?
 - a) 10
 - b) 12
 - c) 13
 - d) 17
11. Which of the following are biodegradable pollutants?
 - a) Plastics
 - b) Domestic sewage
 - c) Detergent
 - d) All of these
12. BOD means
 - a) Biochemical Oxygen Demand
 - b) Chemical Oxygen Demand
 - c) Biophysical oxygen Demand
 - d) All of these
13. Sound beyond which of the following level can be regarded as a pollutant.
 - a) 40 dB
 - b) 80 dB
 - c) 120 dB
 - d) 150 dB
14. Which of the following is a point source of water pollution?
 - a) Factories
 - b) Sewage treatment plants
 - c) Urban & Sub – urban lands
 - d) Both (a) & (b)
15. Pesticide causes
 - a) Eye irritation
 - b) Skin irritation
 - c) Respiratory ailment
 - d) All of these
16. Amount of waste infectious produced in hospitals.
 - a) 45 %
 - b) 65 %
 - c) 80 %
 - d) 100 %
17. Cytotoxic and expired drugs are disposed of by
 - a) Dumping
 - b) Autoclave
 - c) Incineration
 - d) All of these
18. Color code of plastic bag for disposing of microbial laboratory culture waste
 - a) Black
 - b) Red
 - c) Blue
 - d) White
19. All of the waste are incinerated except
 - a) Reactive chemical waste
 - b) Vaccine
 - c) Mutilated parts
 - d) Discarded drugs
20. The average composition of municipal solid waste is
 - a) 41 % organic , 40 % inert & 19 % recyclable.
 - b) 20 % organic , 60% inert & 20 % recyclable
 - c) 30 % organic , 20% inert & 50 % recyclable
 - d) 19 % organic , 41% inert & 40 % recyclable
21. Major atmospheric gas layer in stratosphere is
 - a) Hydrogen
 - b) Carbon dioxide
 - c) Ozone
 - d) Helium
22. World Environment day is on ____
 - a) 5th May
 - b) 5th June
 - c) 3rd Jan
 - d) 1st June

23. Which of the following is not a green house gas?
a) Hydro chlorofluorocarbons b) Methane
c) CO₂ d) SO₂
24. Global warming could affect
a) Climate b) Increase in sea level
c) Melting of glaciers d) All of these
25. Primary cause of acid rain around the world is due to
a) Carbon dioxide b) Sulphur dioxide c) Carbon monoxide d) Ozone
26. What is the main reason for the depletion of groundwater levels?
a) Rains are less frequent
b) Tree cover has reduced
c) Tube wells and hand pumps are used for irrigation
d) The course of river being changed and dams built
27. What happens when water contains 8 – 20 PPM of fluoride concentration?
a) Blue baby disease b) Crippling fluorosis
c) Dental fluorosis d) Mottling of teeth
28. Coral reefs of India are located in
a) Goa b) Himalayan region
c) Andaman & Nicobar Islands d) Uttar Pradesh
29. What is the allowable concentration of fluorides in drinking water?
a) 1.0 mg/L b) 4.25 mg/L c) 1.5 mg/L d) 1.75 mg/L
30. Minmata disease is caused by
a) Lead b) Mercury c) Cadmium d) Arsenic
31. When did National Disaster Management Authority form?
a) 2000 b) 2005 c) 2010 d) 2015
32. Scientific study of earthquake is called
a) Seismograph b) Seismology c) Both (a) & (b) d) None of these
33. Mass killing diseases can be referred as
a) Biological disaster b) Industrial disaster
c) War disaster d) Flood disaster
34. South Africa is leading exporter of which mineral?
a) Copper b) Diamond c) Silver d) Gold
35. Which one of the following is an exhaustible resource?
a) Coal b) Rain precipitation c) Air d) Tidal energy
36. Which of the following is a non – renewable resource?
a) Solar energy b) Wind energy c) Fossil fuels d) Hydro power

37. Which of the following is an example of sustainable use of water resources?
 a) Overuse of ground water b) Building dams on rivers
 c) Rainwater harvesting d) Dumping industrial waste in rivers
38. Geothermal energy is a
 a) Heat energy b) Current energy c) Wind energy d) Solar energy
39. Molasses from sugar industry is used to generate
 a) Biodiesel b) Hydrogen c) Bio-ethanol d) Bio - methanol
40. Nuclear power plant in Karnataka is located at
 a) Bhadravati b) Sandur c) Raichur d) Kaiga
41. Wild Life Protection Act in India was passed in
 a) 1978 b) 1972 c) 1986 d) 1992
42. Environment Protection Act was enacted in year
 a) 1986 b) 1974 c) 1992 d) 1984
43. Water Protection Act was enacted in
 a) 1974 b) 1990 c) 1985 d) 2021
44. Blue baby syndrome is caused by the contamination of water due to
 a) Phosphates b) Sulphur c) Arsenic d) Nitrates
45. Which toxic compound is not found in e – waste?
 a) Mercury b) Cadmium c) Neon d) Lead
46. What are the health hazards caused by E – waste?
 a) Lung cancer b) DNA damage c) Brain d) All of these
47. The Kyoto protocol was adopted at the
 a) 3rd conference of UNFCCC in 1977
 b) Convention on the trans boundary effects of industrial accidents
 c) UNFCCC in 1992
 d) Convention on Biological diversity
48. The primary cause of acid rain around the world is due to
 a) Carbon dioxide b) Sulphur dioxide c) Carbon monoxide d) Ozone
49. The carbon “credit is permit” representing the right to emit.
 a) One tonne of CO₂ b) 10 tonnes of CO₂
 c) 5 tonnes of CO₂ d) 15 tonnes of CO₂
50. Fossil fuels are converted into energy by
 a) Burning b) Cooling c) Sublimation d) Melting

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